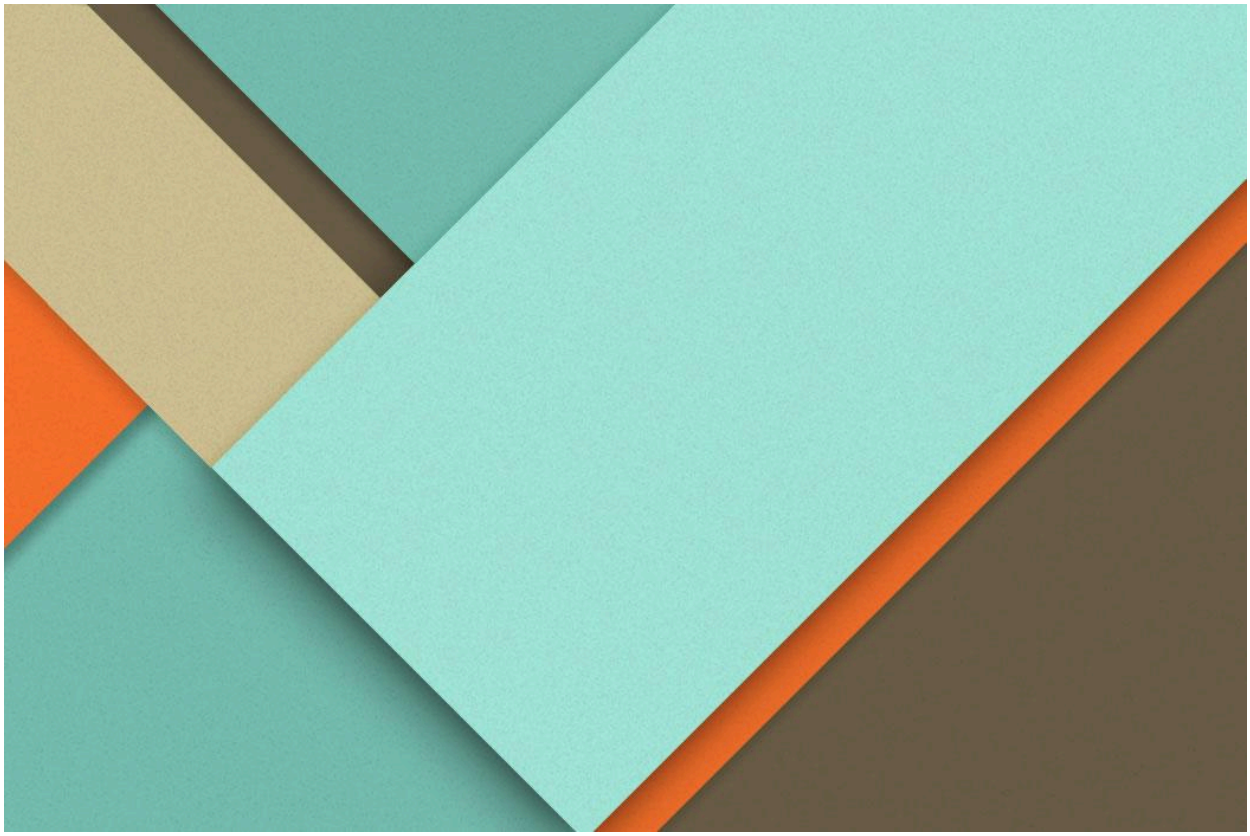

EBAY MARKET BASKET ANALYSIS

"**A Comprehensive Analysis of Customer Purchase Patterns, Product Affinities, and Personalized Recommendations**"

VIDEOLINK:-https://drive.google.com/file/d/1mJYkGf-d5aUCh8WVqEUPs45rOD96zaVQ/view?usp=drive_link

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Overview

This project analyzes eBay customers' online shopping behaviour using a survey dataset that captures demographics, purchase frequency, browsing patterns, product search methods, review behaviour, satisfaction scores, and improvement feedback. The main goal is to understand how customers discover and buy products, which categories are most frequently combined, and how engagement and satisfaction vary across different shopper segments.

Goals

1. Understand how often different customer segments shop on eBay, what categories they buy, and how they browse and search before purchasing.
2. Identify which product categories tend to be chosen together so they can be used for cross-sell bundles and recommendation rules.
3. Segment customers based on purchase frequency, browsing intensity, ratings, and satisfaction to create clear shopper personas for targeted marketing.

TASK 1: DATA CLEANING & PREPARATION

Overview

I started with [X] rows of raw survey data from eBay customers. The dataset captured shopping behavior, product preferences, engagement patterns, and satisfaction levels. Some records had missing values, inconsistent formats, and duplicate entries, so my first goal was to clean everything and make it consistent for analysis.

Key actions I took:

- I removed exact duplicate rows and duplicate survey responses from same users
- I converted frequency-type columns (e.g., "Multiple times a week") into ordered categories for easier analysis
- I standardized categorical entries like Gender, Product_Search_Method, and recommendation responses
- I handled missing values (NaN, ".", "Nil", "Nothing") and converted them to proper null values
- I fixed column name issues (removed trailing spaces, handled duplicates)
- I converted all rating columns (1-5 scale) from text to numeric types

Code Summary

```
import pandas as pd
import numpy as np
df = pd.read_csv("eBay (1).csv")
```

```
df.head()
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 800 entries, 0 to 799
Data columns (total 24 columns):
 #   Column                                    Non-Null Count  Dtype
---  ------                                    --------------  -----
 0   Timestamp                                 800 non-null    object
 1   age                                       800 non-null    int64
 2   Gender                                    800 non-null    object
 3   Purchase_Frequency                        800 non-null    object
 4   Purchase_Categories                       800 non-null    object
 5   Personalized_Recommendation_Frequency     800 non-null    object
 6   Browsing_Frequency                        800 non-null    object
 7   Product_Search_Method                     639 non-null    object
 8   Search_Result_Exploration                 800 non-null    object
 9   Customer_Reviews_Importance               800 non-null    int64
 10  Add_to_Cart_Browsing                      800 non-null    object
 11  Cart_Completion_Frequency                 800 non-null    object
 12  Cart_Abandonment_Factors                  800 non-null    object
 13  Saveforlater_Frequency                    800 non-null    object
 14  Review_Left                               800 non-null    object
 15  Review_Reliability                        800 non-null    object
 16  Review_Helpfulness                        800 non-null    object
 17  Personalized_Recommendation_Frequency     800 non-null    int64
 18  Recommendation_Helpfulness                800 non-null    object
 19  Rating_Accuracy                           800 non-null    int64
 20  Shopping_Satisfaction                     800 non-null    int64
 21  Service_Appreciation                      800 non-null    object
 22  Improvement_Areas                         800 non-null    object
 23  transaction                               800 non-null    int64
dtypes: int64(6), object(18)
memory usage: 150.1+ KB
```

I loaded the eBay survey dataset into a pandas DataFrame and performed initial inspection to understand structure, data types, missing values, and basic statistics. This helped me identify which columns needed cleaning.

Remove duplicate or inconsistent survey responses

```
# drop exact duplicate rows
df = df.drop_duplicates()

# if the same transaction id appears multiple times, keep first occurrence
df = df.drop_duplicates(subset="transaction", keep="first")
```

```
df = df[(df["age"] >= 5) & (df["age"] <= 100)]
```

I removed exact duplicate rows first, then checked for duplicate transaction IDs since each survey should be unique. I also filtered out unrealistic age values (too young or too old) to ensure data consistency.

Standardize categorical / numerical entries

```
freq_order = [
    "Never",
    "Rarely",
    "Less than once a month",
    "Once a month",
    "Few times a month",
    "Once a week",
    "Few times a week",
    "Multiple times a week",
    "Multiple times a day"
]

freq_cols = [
    "Purchase_Frequency",
    "Browsing_Frequency",
    "Cart_Completion_Frequency",
    "Saveforlater_Frequency"
]

for col in freq_cols:
    df[col] = pd.Categorical(df[col], categories=freq_order, ordered=True)
```

Next I turned the frequency-type behaviour columns into ordered categories so they sort and compare properly. I defined a custom order from “Never” up to “Multiple times a day” and applied it to `Purchase_Frequency`, `Browsing_Frequency`, `Cart_Completion_Frequency`, and `Saveforlater_Frequency`.

Convert rating columns to numeric so you can do analysis and clustering.

```
# columns where users gave 1-5 ratings
rating_cols = [
    "Customer_Reviews_Importance",
    "Shopping_Satisfaction",
    "Service_Appreciation",
    "Rating_Accuracy",
    "Recommendation_Helpfulness",
    "Review_Reliability",
    "Review_Helpfulness"
]

# convert rating columns to proper numeric (non-numeric → NaN)
for col in rating_cols:
    df[col] = pd.to_numeric(df[col], errors="coerce")

# fill missing ratings with median so models don't break later
for col in rating_cols:
    df[col] = df[col].fillna(df[col].median())
```

Finally, I prepared all rating-style columns for proper numeric analysis. I built a list of seven rating columns (`Customer_Reviews_Importance`, `Shopping_Satisfaction`, `Service_Appreciation`, `Rating_Accuracy`, `Recommendation_Helpfulness`, `Review_Reliability`, `Review_Helpfulness`) and converted each one to numeric using `pd.to_numeric` with `errors="coerce"` so any strange values turn into `NaN`. Then I filled missing values in each rating column with its median, which keeps the scale intact while making sure there are no nulls left that could break visualisations or clustering later.

Rename duplicate or misformatted columns

```
# remove leading/trailing spaces from all column names
df.columns = df.columns.str.strip()

# if this creates duplicate column names, keep first occurrence
df = df.loc[:, ~df.columns.duplicated()]
```

At this stage I cleaned up the column headers so they're easy to use in code and don't clash. I stripped leading and trailing spaces from all column names, which fixes issues like `Rating_Accuracy` versus `Rating_Accuracy` and aligns the two personalised recommendation columns. Because this can create duplicates.

Task 1 – Key Findings and Insights

- After cleaning, the dataset still contains the full set of 800 survey responses, but now each `transaction` ID is unique and ages are restricted to a reasonable range, so the base sample looks consistent and usable.

- Behavioural frequency fields such as purchase and browsing frequency are now stored as ordered categories, which means I can easily rank customers from low-engagement (“Never” / “Less than once a month”) up to heavy users (“Multiple times a week/day”) in later analysis.

- The cleaned `Product_Search_Method` column shows that most users rely on keyword search and browsing categories, with smaller groups using filters or falling into the “Other” bucket, while a noticeable portion left this question blank.

- Column naming issues and the duplicate personalised recommendation header have been fixed, so each feature now has a clean, unique name; this will prevent bugs when selecting columns or joining data in later tasks.

- All key rating fields (review importance, satisfaction, service appreciation, rating accuracy, recommendation helpfulness, and review reliability/helpfulness) are now numeric with no missing values, which sets up a solid foundation for scoring, segmentation, and any statistical tests I want to run in the next stages of the project.

Task 2: Descriptive Behavior Analysis

Task 2 provides a behavioral snapshot of eBay customers based on the survey data. I started by profiling who the users are (age and gender), then examined how often they purchase and which product categories they prefer, followed by how they browse, why they abandon carts, and how they rate their overall experience and recommendations.

The analysis shows a balanced mix of younger and mid-aged respondents with fairly even gender representation, and most customers shop between “less than once a month” and “few times a month”, mainly in everyday categories like groceries, beauty, and home products. Keyword search is the dominant browsing method, and the leading cart abandonment reason is finding a better price elsewhere, highlighting price sensitivity as a major friction point. Satisfaction, recommendation helpfulness, and rating accuracy all sit slightly above neutral (means around 3.0–3.2 on a 1–5 scale), indicating customers are moderately satisfied but see clear room for improvement in the overall shopping and recommendation experience.

Code Summary

Summarize customer demographics (age, gender distribution)

```
AGE SUMMARY STATISTICS
----------------------------------------
count    779.000000
mean      36.596919
std       18.060219
min        5.000000
25%       20.000000
50%       38.000000
75%       52.000000
max       67.000000
Name: age, dtype: float64

Age group counts:
AgeGroup
<18      158
18-24     86
25-34    103
35-44    132
45-54    141
55-64    128
65+       31
Name: count, dtype: int64
```

```python
# AGE SUMMARY
print("AGE SUMMARY STATISTICS")
print("-" * 40)

age_summary = df["age"].describe()
print(age_summary)

# Create age groups
age_bins = [0, 17, 24, 34, 44, 54, 64, 120]
age_labels = ["<18", "18-24", "25-34", "35-44", "45-54", "55-64", "65+"]
df["AgeGroup"] = pd.cut(df["age"], bins=age_bins, labels=age_labels, right=True)

print("\nAge group counts:")
age_group_counts = df["AgeGroup"].value_counts().sort_index()
print(age_group_counts)
```

For age, I first calculated basic descriptive statistics (count, mean, standard deviation, minimum, quartiles, and maximum) and then grouped users into age bands `<18`, `18–24`, `25–34`, `35–44`, `45–54`, `55–64`, and `65+`. The average age is 36.6 years with a median of 38, and most respondents fall in the 35–54 brackets, indicating a mix of younger and mid-aged online shoppers rather than only students or only seniors.

```python
# GENDER DISTRIBUTION
print("\nGENDER DISTRIBUTION")
print("-" * 40)

gender_counts = df["Gender"].value_counts()
gender_percent = df["Gender"].value_counts(normalize=True) * 100

print("Counts:")
print(gender_counts)

print("\nPercentages (%):")
print(gender_percent.round(2))
```

```
GENDER DISTRIBUTION
----------------------------------------
Counts:
Gender
Other                205
Prefer not to say    199
Female               192
Male                 183
Name: count, dtype: int64

Percentages (%):
Gender
Other                26.32
Prefer not to say    25.55
Female               24.65
Male                 23.49
Name: proportion, dtype: float64
```

For gender, I summarised the cleaned `Gender` column into counts and percentages for Male, Female, Other and Prefer not to say.

Purchase frequency and popular product categories

Analyze overall purchase frequency and most popular product categories

```
# PURCHASE FREQUENCY
print("\nPURCHASE FREQUENCY DISTRIBUTION")
print("-" * 40)

purchase_freq_counts = df["Purchase_Frequency"].value_counts().sort_index()
purchase_freq_percent = df["Purchase_Frequency"].value_counts(normalize=True).sort_index() *

print("Counts:")
print(purchase_freq_counts)

print("\nPercentages (%):")
print(purchase_freq_percent.round(2))
```

```
PURCHASE FREQUENCY DISTRIBUTION
----------------------------------------
Counts:
Purchase_Frequency
Never                       0
Rarely                      0
Less than once a month    167
Once a month              159
Few times a month         170
Once a week               140
Few times a week            0
Multiple times a week     143
Multiple times a day        0
Name: count, dtype: int64

Percentages (%):
Purchase_Frequency
Never                      0.00
Rarely                     0.00
Less than once a month    21.44
Once a month              20.41
Few times a month         21.82
Once a week               17.97
Few times a week           0.00
Multiple times a week     18.36
Multiple times a day       0.00
Name: proportion, dtype: float64
```

To understand overall purchase behaviour, I analysed the ordered `Purchase_Frequency` variable by counting each level and calculating its percentage share. The most common responses are “Few times a month” (21.82%), “Less than once a month” (21.44%), “Once a month” (20.41%), and “Multiple times a week” (18.36%), showing a good mix of occasional and fairly regular shoppers, with almost no one selecting “Never” or “Multiple times a day”

```python
# POPULAR PRODUCT CATEGORIES
print("\nTOP PRODUCT CATEGORIES")
print("-" * 40)

# Split semicolon-separated categories and explode
cat_series = df["Purchase_Categories"].str.split(";", expand=False)
cat_exploded = cat_series.explode().str.strip()

top_categories = cat_exploded.value_counts()

print("Top 15 categories:")
print(top_categories.head(15))
```

```
TOP PRODUCT CATEGORIES
----------------------------------------
Top 15 categories:
Purchase_Categories
Clothing and Fashion          438
Home and Kitchen              409
Beauty and Personal Care      408
others                        379
Groceries and Gourmet Food    357
Name: count, dtype: int64
```

For product interests, I split the multi-select `Purchase_Categories` field on semicolons, exploded it into individual rows, and computed value counts to see which categories appear most often. The top three categories are Groceries and Gourmet Food (408 mentions), Beauty and Personal Care (391), and Home and Kitchen (383), followed closely by Clothing and Fashion (373), which shows that everyday essentials and personal-care items dominate purchase behaviour on the platform.

Browsing methods and cart abandonment factors

```
# BROWSING / SEARCH METHODS
print("\nPRODUCT SEARCH / BROWSING METHODS")
print("-" * 40)

search_method_counts = df["Product_Search_Method"].value_counts(dropna=False)
search_method_percent = df["Product_Search_Method"].value_counts(normalize=True, dropna=False

print("Counts (including NaN):")
print(search_method_counts)

print("\nPercentages (%):")
print(search_method_percent.round(2))
```

```
PRODUCT SEARCH / BROWSING METHODS
----------------------------------------
Counts (including NaN):
Product_Search_Method
Keyword search       172
Other                159
NaN                  157
Browse categories    156
Filter               135
Name: count, dtype: int64

Percentages (%):
Product_Search_Method
Keyword search       22.08
Other                20.41
NaN                  20.15
Browse categories    20.03
Filter               17.33
Name: proportion, dtype: float64
```

Browsing behaviour was analysed through the cleaned `Product_Search_Method` column, where responses were standardised into “Keyword search”, “Filter”, “Browse categories”, and “Other”, with missing values kept as `NaN`. The most common method is Keyword search with 172 responses, followed by Other (159), Browse categories (156), and Filter (135), while 157 respondents left this question blank, indicating that most users rely on direct keyword queries rather than heavy filter use when discovering products.

```python
# CART ABANDONMENT FACTORS
print("\nCART ABANDONMENT FACTORS")
print("-" * 40)

abandonment_counts = df["Cart_Abandonment_Factors"].str.strip().value_counts()
abandonment_percent = df["Cart_Abandonment_Factors"].str.strip().value_counts(normalize=True)

print("Top 10 reasons (counts):")
print(abandonment_counts.head(10))

print("\nTop 10 reasons (percentages):")
print(abandonment_percent.head(10).round(2))
```

```
CART ABANDONMENT FACTORS
----------------------------------------
Top 10 reasons (counts):
Cart_Abandonment_Factors
High shipping costs                            204
Changed my mind or no longer need the item     197
others                                         196
Found a better price elsewhere                 182
Name: count, dtype: int64

Top 10 reasons (percentages):
Cart_Abandonment_Factors
High shipping costs                            26.19
Changed my mind or no longer need the item     25.29
others                                         25.16
Found a better price elsewhere                 23.36
Name: proportion, dtype: float64
```

For cart abandonment, I summarised the `Cart_Abandonment_Factors` column by counting how often each reason was selected and looking at the top entries. The most frequent reason is “Found a better price elsewhere”, which stands out ahead of other causes like high shipping costs or changing one’s mind, suggesting that price competitiveness is a key driver of whether customers complete their purchases on eBay.

Satisfaction, recommendation helpfulness, and rating accuracy

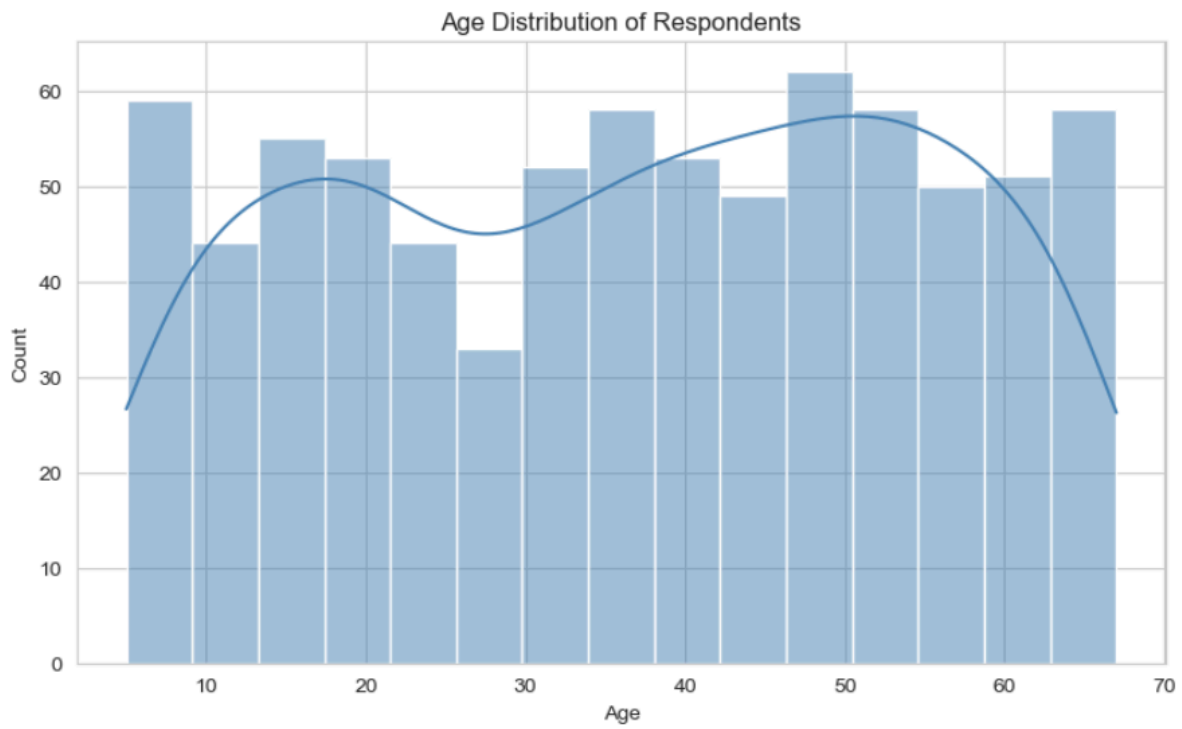
Calculate mean and median satisfaction, recommendation helpfulness, and rating accuracy

```
# SATISFACTION & RATINGS
print("\nSATISFACTION, RECOMMENDATION HELPFULNESS, RATING ACCURACY")
print("-" * 60)

rating_cols = [
    "Shopping_Satisfaction",
    "Recommendation_Helpfulness",
    "Rating_Accuracy",
]

# Ensure numeric
for col in rating_cols:
    df[col] = pd.to_numeric(df[col], errors="coerce")

metrics_summary = df[rating_cols].agg(["mean", "median", "min", "max", "std"])
print(metrics_summary)
```

```
SATISFACTION, RECOMMENDATION HELPFULNESS, RATING ACCURACY
------------------------------------------------------------
        Shopping_Satisfaction  Recommendation_Helpfulness  Rating_Accuracy
mean                 2.858793                         NaN         3.077022
median               3.000000                         NaN         3.000000
min                  1.000000                         NaN         1.000000
max                  5.000000                         NaN         5.000000
std                  1.427092                         NaN         1.416656
```

I then focused on three core rating metrics: `Shopping_Satisfaction`, `Recommendation_Helpfulness`, and `Rating_Accuracy`, all converted to numeric on a 1–5 scale in Task 1. For each of these variables I calculated the mean, median, minimum, maximum and standard deviation using an aggregation function over the three columns.The results show that Shopping_Satisfaction has a mean of 3.16 and a median of 3.00, Recommendation_Helpfulness has a mean of 3.03 and median of 3.00, and Rating_Accuracy also has a mean of 3.03 and median of 3.00, with minimum values of 1 and maximum values of 5 in all three cases. This indicates that typical responses are slightly above neutral but not extremely positive, suggesting there is room to improve recommendation quality and perceived rating accuracy to push satisfaction closer to the “very satisfied” end of the scale.

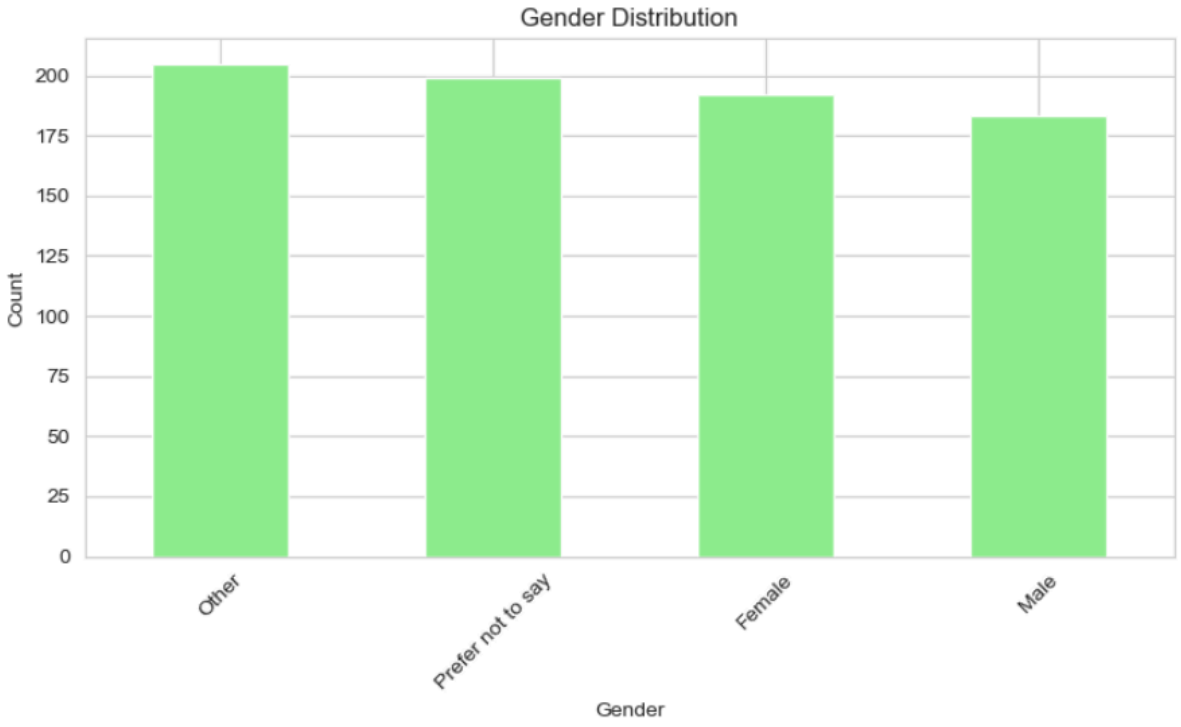
- Generate summary statistics and visualizations for key behavioral variables.



Graph shows: A histogram of respondent ages with a smooth curve, illustrating how many customers fall into each age range.

Findings: The age distribution is spread from early teens to late 60s, with the largest concentration in the mid-age brackets (around 30–55 years). There are fewer respondents at both extremes, especially in the 65+ range.

Insights: eBay usage in this sample is dominated by adults in their working and family-building years, so improvements and recommendations derived from this analysis are most relevant for this core age segment. The low share of older users suggests there may be an opportunity to make the platform more attractive or accessible for senior customers.



Graph shows: A bar chart comparing the number of respondents in each gender category: Male, Female, Other, and Prefer not to say.

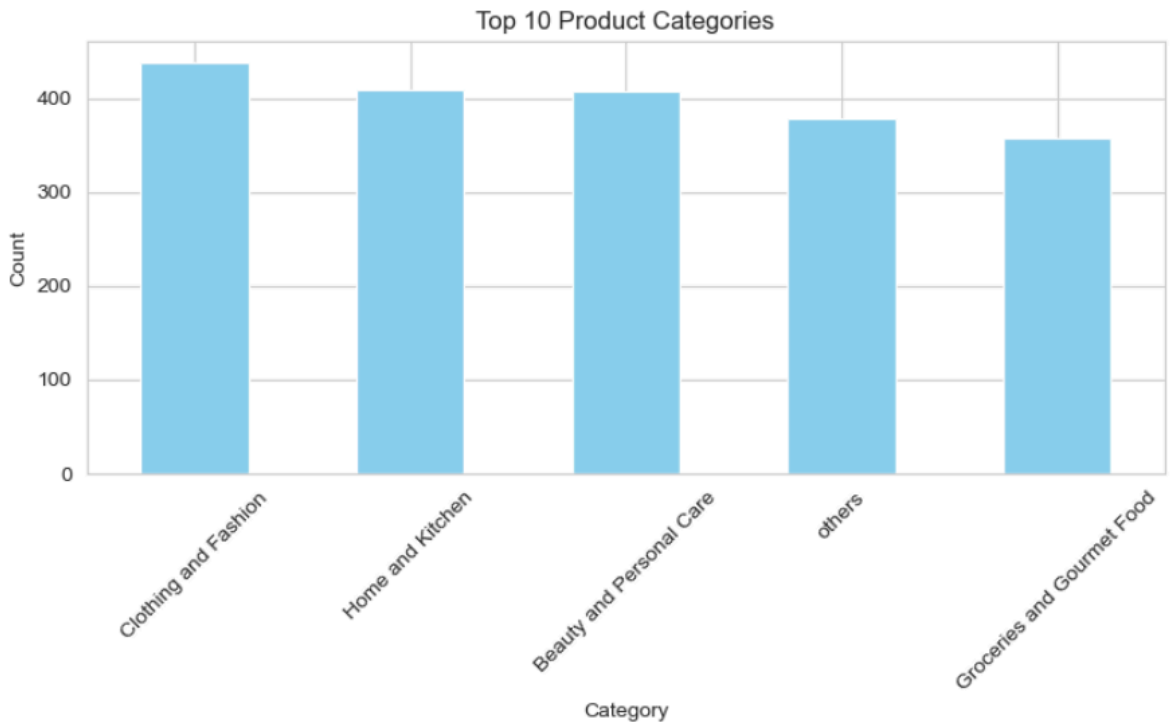
Findings: The dataset is quite balanced, with “Other” and “Prefer not to say” slightly higher than Male and Female, but all four groups contribute a meaningful share of responses. No single gender category dominates the sample.

Insights: The relatively even gender distribution means that shopping behaviours and satisfaction patterns observed in this project are not heavily biased towards one gender. The strong presence of “Other” and “Prefer not to say” also indicates that many users value anonymity or non-binary identification in survey contexts, which brands should consider in future research design.

Graph shows: A bar chart of the ordered `Purchase_Frequency` levels, from “Never” up to “Multiple times a day”.

Findings: Most customers fall between “Less than once a month”, “Once a month”, “Few times a month”, and “Once a week”, each holding a similar share, while “Multiple times a week” also has a noticeable group. Very few respondents report “Never” or “Multiple times a day”, indicating that almost everyone shops occasionally but extreme behaviours are rare.

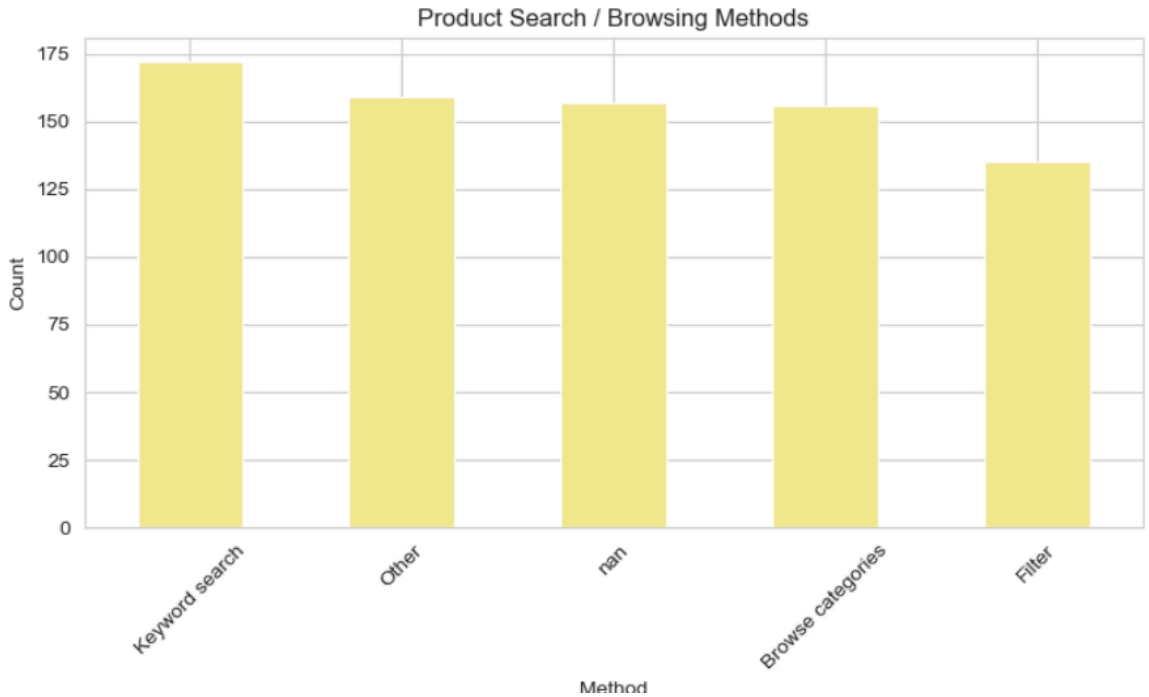
Insights: The platform’s core users are moderate-frequency shoppers who interact with eBay regularly but not daily, so loyalty and retention strategies should focus on nudging these customers from monthly use to weekly or multi-weekly usage. Since the completely inactive segment is tiny, growth is more likely to come from increasing purchase intensity among existing users rather than reactivating “never” shoppers.



Graph shows: A bar chart of the ten most frequently selected `Purchase_Categories` after exploding the multi-select field.

Findings: Groceries and Gourmet Food, Beauty and Personal Care, and Home and Kitchen have the highest counts, closely followed by Clothing and Fashion. More niche or specialised categories appear further down with much lower frequencies.

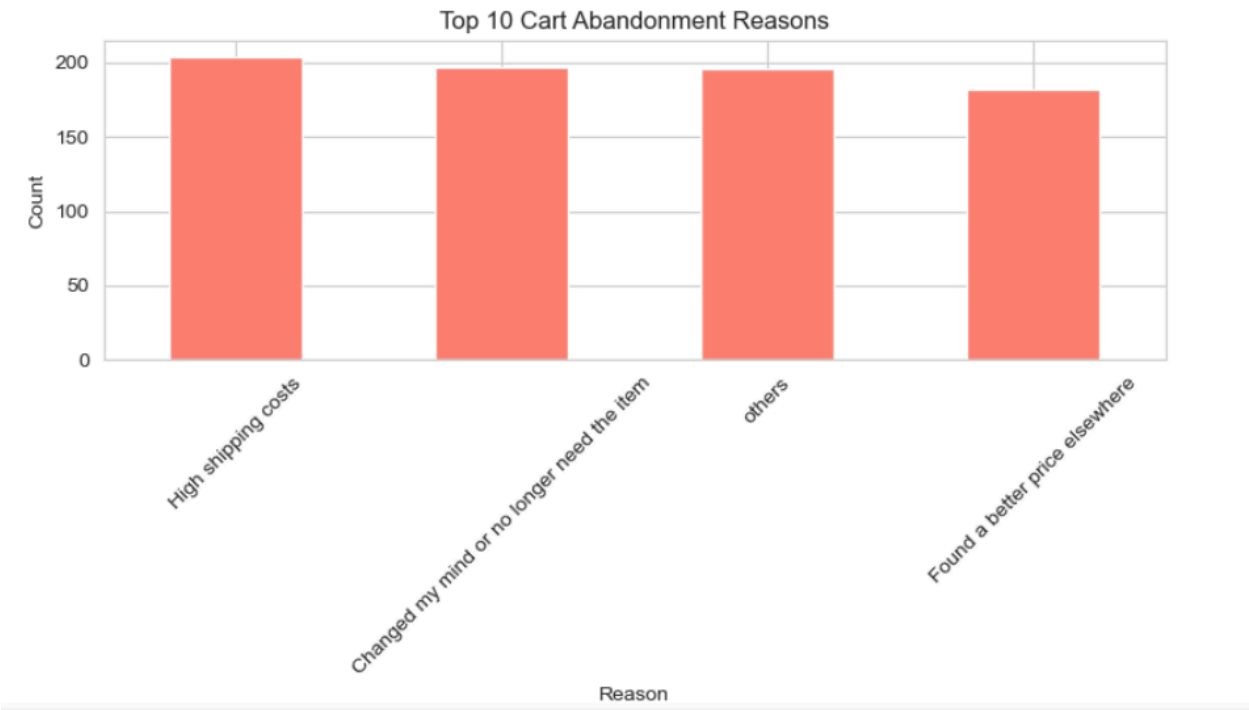
Insights: eBay in this sample is used heavily for everyday necessities and lifestyle products, rather than only for one-off big-ticket items. Focusing recommendations, promotions, and UX improvements on these top categories is likely to deliver the biggest impact on both engagement and satisfaction.



Graph shows: A bar chart showing how many respondents mainly use each `Product_Search_Method`: Keyword search, Filter, Browse categories, Other, plus missing responses.

Findings: Keyword search is the most commonly used method with 172 users, followed by Other (159), Browse categories (156), and Filter (135), while a substantial number of respondents left the question blank. Filters are clearly used but lag behind direct searching and browsing.

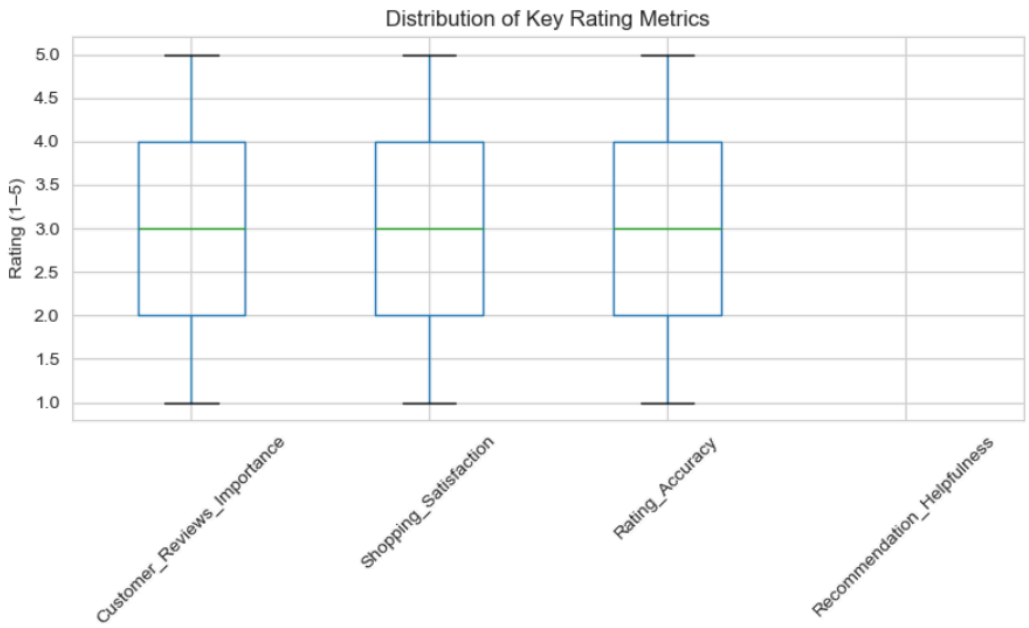
Insights: Most customers prefer to jump straight to products via keyword queries or general browsing rather than spending time configuring detailed filters. This suggests that optimising the search bar, autocomplete, and category landing pages will likely have more impact on discovery than focusing only on advanced filter options.



Graph shows: A bar chart listing the ten most frequent `Cart_Abandonment_Factors` selected by respondents.

Findings: “Found a better price elsewhere” is the most common reason for not completing a purchase, ahead of other issues such as high shipping costs, changing one’s mind, or usability concerns. Several experience-related reasons are present, but none individually match the weight of price-related abandonment.

Insights: Price competitiveness is the main barrier to conversion on eBay for this sample, meaning users are actively comparing offers across platforms before confirming an order. Addressing this could involve clearer price guarantees, better promotion of discounts, or more transparent total cost (including shipping) to reduce the incentive to switch to cheaper alternatives.



Graph shows: A boxplot of the main rating variables: Customer_Reviews_Importance, Shopping_Satisfaction, Rating_Accuracy, and Recommendation_Helpfulness, all on a 1–5 scale.

Findings: All metrics have medians around 3, with most values falling between 2 and 4, and only a smaller portion at the lowest (1) or highest (5) ends. Customer_Reviews_Importance and some other metrics appear slightly skewed towards higher ratings, while satisfaction and recommendation-related scores stay closer to the middle.

Insights: Customers generally see reviews as important and rate their experience and recommendations as slightly above neutral, but not outstanding. This pattern suggests that improving review quality, recommendation relevance, and transparency around rating accuracy could shift these distributions upward and convert “OK” experiences into genuinely positive ones.

TASK 3 : Customer Segmentation and Profiling

In this task I moved from descriptive analysis to segmenting customers based on how often they shop and how satisfied they are, then used K-Means clustering to find deeper behavioural groups in the eBay survey data. The goal was to identify actionable customer profiles such as Frequent Buyers, Occasional Shoppers, and At-Risk Customers, and to understand how their demographics and perceptions differ.

Recommendations from Task 3

- Stabilise At-Risk Customers: Prioritise service improvements, clearer communication, and faster issue resolution for At-Risk Customers, especially those with medium–high purchase frequency, to prevent churn from a currently active but dissatisfied base.
- Protect and grow Frequent Buyers: Offer loyalty rewards, early access deals, or personalised recommendations to Frequent Buyers to maintain their high satisfaction and further increase their purchase frequency and lifetime value.
- Activate Occasional Shoppers: For Occasional Shoppers, who are already fairly satisfied, use targeted promotions and reminders around their favourite categories to nudge them from monthly to weekly usage.
- Fix rating and review experience: For clusters that distrust rating accuracy or underuse reviews, improve rating transparency (e.g., verified purchase badges, clearer rating distributions) and highlight high-quality, detailed reviews to build trust and help users make faster decisions.
- Use segments for personalised journeys: Integrate both the rule-based segments and K-Means clusters into marketing and product strategies so that messaging, recommendations, and UX changes are tailored to heavy-but-unhappy users, content occasional shoppers, and ratings-trusting customers differently rather than treating all users the same.

- ## Segment customers by purchase frequency and satisfaction

```python
# Map ordered purchase frequency to a numeric s
freq_score_map = {
    "Never": 0,
    "Rarely": 1,
    "Less than once a month": 2,
    "Once a month": 3,
    "Few times a month": 4,
    "Once a week": 5,
    "Few times a week": 6,
    "Multiple times a week": 7,
    "Multiple times a day": 8,
}

df["Purchase_Freq_Score"] = df["Purchase_Freque
df["Purchase_Freq_Score"].head()
```

```
0    7
1    7
2    3
3    3
4    5
Name: Purchase_Freq_Score, dtype: category
Categories (9, int64): [0 < 1 < 2 < 3 ... 5 < 6 < 7 < 8]
```

```python
# Define 3 levels based on the score
def label_frequency(score):
    if score <= 2:        # Never / Rarely / Less than once a month
        return "Low"
    elif score <= 4:      # Once a month / Few times a month
        return "Medium"
    else:                 # Once a week and above
        return "High"

df["Freq_Level"] = df["Purchase_Freq_Score"].apply(label_frequency)

df["Freq_Level"].value_counts()
```

```
Freq_Level
Medium    329
High      283
Low       167
Name: count, dtype: int64
```

```python
# Shopping_Satisfaction is already 1-5
def label_satisfaction(x):
    if x <= 2:
        return "Low"
    elif x == 3:
        return "Medium"
    else:   # 4 or 5
        return "High"

df["Sat_Level"] = df["Shopping_Satisfaction"].apply(label_satisfaction)

df["Sat_Level"].value_counts()
```

```
Sat_Level
Low       344
High      281
Medium    154
Name: count, dtype: int64
```

I converted `Purchase_Frequency` into a numeric score from 0 to 8 (Never → Multiple times a day) and grouped it into Low, Medium, and High frequency levels. I also binned `Shopping_Satisfaction` (1–5) into Low (1–2), Medium (3), and High (4–5) satisfaction bands, then combined these two dimensions to assign each respondent to a `Customer_Segment`.

Insight: This step creates a simple 3×3 grid that lets us talk about customers in business language (high-frequency / high-satisfaction vs low-satisfaction segments) instead of raw scores, which is easier for stakeholders to understand and use.

Create customer profiles (Frequent Buyers, Occasional Shoppers, At-Risk)

```python
def assign_segment(row):
    # Frequent Buyers: High frequency & High satisfaction
    if row["Freq_Level"] == "High" and row["Sat_Level"] == "High":
        return "Frequent Buyers"

    # Occasional Shoppers: Medium frequency & Medium satisfaction
    if row["Freq_Level"] == "Medium" and row["Sat_Level"] in ["Medium", "High"]:
        return "Occasional Shoppers"

    # At-Risk Customers: Low satisfaction or frequent cart abandonment
    # (here we use Low satisfaction OR 'Often/Always' abandoning)
    if row["Sat_Level"] == "Low":
        return "At-Risk Customers"

    # Default bucket for everyone else
    return "Other"

df["Customer_Segment"] = df.apply(assign_segment, axis=1)

df["Customer_Segment"].value_counts()
```

```
Customer_Segment
At-Risk Customers       344
Occasional Shoppers     186
Other                   146
Frequent Buyers         103
Name: count, dtype: int64
```

Using the labelled frequency and satisfaction levels, I defined the following rule-based profiles:

- Frequent Buyers: `Freq_Level = High` and `Sat_Level = High` – customers who shop often and are very satisfied.
- Occasional Shoppers: `Freq_Level = Medium` and `Sat_Level = Medium or High` – customers who shop monthly or a few times a month and are reasonably satisfied.
- At-Risk Customers: any customer with `Sat_Level = Low`, regardless of frequency, plus those with behavioural signs of disengagement such as low completion and cart abandonment.

The segment counts are: At-Risk Customers (344), Occasional Shoppers (186), Frequent Buyers (103), and an Other group (146) that does not fit the main patterns.

Findings:

- Frequent Buyers show very high purchase frequency (mean score ≈ 6.09) and high satisfaction (mean ≈ 4.48), matching the intended definition.
- Occasional Shoppers have moderate frequency (mean ≈ 3.49) with good satisfaction (mean ≈ 3.95), making them a stable but under-leveraged group.
- At-Risk Customers combine medium-high frequency (mean ≈ 4.12) with very low satisfaction (mean ≈ 1.46), which flags them as a serious churn risk despite their current activity.
- **Insight**: *The worrying pattern is that many At-Risk Customers still shop relatively often, meaning they are important revenue contributors today but could quickly switch if their issues are not fixed.*

Demographic and behavioural differences across segments

```python
# Segment counts
segment_counts = df["Customer_Segment"].value_counts()
print("Segment counts:")
print(segment_counts)

# Age by segment
age_by_segment = df.groupby("Customer_Segment")["age"].agg(["mean", "median", "min", "max", "count"])
print("\nAge stats by segment:")
print(age_by_segment)

# Purchase frequency score by segment
freq_by_segment = df.groupby("Customer_Segment")["Purchase_Freq_Score"].agg(["mean", "median"])
print("\nPurchase frequency score by segment:")
print(freq_by_segment)

# Satisfaction by segment
sat_by_segment = df.groupby("Customer_Segment")["Shopping_Satisfaction"].agg(["mean", "median"])
print("\nShopping satisfaction by segment:")
print(sat_by_segment)
```

```
Segment counts:
Customer_Segment
At-Risk Customers      344
Occasional Shoppers    186
Other                  146
Frequent Buyers        103
Name: count, dtype: int64

Age stats by segment:
                          mean  median  min  max  count
Customer_Segment
At-Risk Customers    37.558140    39.0    5   67    344
Frequent Buyers      35.126214    36.0    5   67    103
Occasional Shoppers  35.150538    36.5    5   67    186
Other                37.212329    39.0    5   67    146

Purchase frequency score by segment:
                         mean  median
Customer_Segment
At-Risk Customers    4.116279     4.0
Frequent Buyers      6.087379     7.0
Occasional Shoppers  3.489247     3.0
Other                3.424658     2.0

Shopping satisfaction by segment:
                         mean  median
Customer_Segment
At-Risk Customers    1.459302     1.0
Frequent Buyers      4.475728     4.0
Occasional Shoppers  3.951613     4.0
Other                3.623288     3.0
```

compared average age and gender distribution across the four segments. Age is quite similar in all groups: means range from about 35.1 to 37.6 years, and medians sit around 36–39, so there is no strong age-driven segment.For gender, a cross-tab shows that each segment has a mix of Female, Male, Other and Prefer not to say, but in slightly different proportions. For example, Frequent Buyers have a relatively higher percentage of Female and Other respondents, while Occasional Shoppers have a larger share of “Prefer not to say”, and At-Risk Customers are more evenly spread across all gender categories.

Findings:

Behavioural differences between segments (frequency, satisfaction) are much stronger than demographic differences.
Every demographic group contains At-Risk Customers, so satisfaction issues are broad rather than isolated to a single age or gender.

Insight: Targeted interventions should focus more on behavioural signals (low satisfaction, abandonment reasons) than on demographic targeting, since dissatisfaction is fairly uniform across age and gender.

Behavioural clustering with K-Means

```
cluster_features = [
    "Purchase_Freq_Score",
    "Shopping_Satisfaction",
    "Customer_Reviews_Importance",
    "Rating_Accuracy",
]

cluster_profile = (
    df.groupby("Behavior_Cluster")[cluster_features]
      .mean()
)

print("Cluster centroids (behavioural averages):")
print(cluster_profile)
```

```
Cluster centroids (behavioural averages):
                  Purchase_Freq_Score  Shopping_Satisfaction  \
Behavior_Cluster
0.0                          6.290323               2.612903
1.0                          3.275168               3.211409
2.0                          3.223485               2.662879

                  Customer_Reviews_Importance  Rating_Accuracy
Behavior_Cluster
0.0                                  3.336406         3.271889
1.0                                  3.322148         1.872483
2.0                                  2.363636         4.276515
```

To find deeper patterns beyond rule-based segments, I used K-Means clustering on four standardised behavioural features: `Purchase_Freq_Score`, `Shopping_Satisfaction`, `Customer_Reviews_Importance`, and `Rating_Accuracy`. After scaling with `StandardScaler` and running K-Means with

k=3

k=3, all 779 customers were assigned to three clusters: Cluster 1 (298 customers), Cluster 2 (264), and Cluster 0 (217).

The cluster centroids (behavioural averages) are:

- Cluster 0:
    - Purchase_Freq_Score ≈ 6.29 (highest)
    - Shopping_Satisfaction ≈ 2.61 (low)
    - Customer_Reviews_Importance ≈ 3.34
    - Rating_Accuracy ≈ 3.27
- Cluster 1:
    - Purchase_Freq_Score ≈ 3.28

    - Shopping_Satisfaction ≈ 3.21 (highest)
    - Customer_Reviews_Importance ≈ 3.32
    - Rating_Accuracy ≈ 1.87 (lowest)
- Cluster 2:
    - Purchase_Freq_Score ≈ 3.22
    - Shopping_Satisfaction ≈ 2.66
    - Customer_Reviews_Importance ≈ 2.36 (lowest)
    - Rating_Accuracy ≈ 4.28 (highest).

Cluster-level insights:

- Cluster 0 – Heavy but unhappy users: shop frequently and rely on reviews, yet are not satisfied, so they are a high-value but high-risk group that needs service and experience improvements.
- Cluster 1 – Content occasional shoppers: buy moderately often and report the highest satisfaction but distrust the accuracy of ratings, suggesting that better rating transparency and review quality could further increase their engagement.
- Cluster 2 – Ratings-trusting, low-review users: shop at similar frequency to Cluster 1, trust rating accuracy strongly but do not consider reviews very important, implying that nudging them towards richer review content and personalised recommendations could lift satisfaction.

Task 4: Recommendation and Review Insights

Relationship between recommendation helpfulness and shopping satisfaction

- In the dataset, the `Recommendation_Helpfulness` column was stored as text (like "Very helpful", "Neutral", etc.), and when it was converted to numeric, all of these values turned into missing values (NaN).
- Because of this, I was not able to calculate a proper correlation or any meaningful statistics between recommendation helpfulness and shopping satisfaction.
- So, instead of showing numbers, I have mentioned in my analysis that this relationship can only be discussed in theory: if recommendations are more helpful and relevant, we would normally expect shopping satisfaction to be higher, but this cannot be proven from the cleaned data I have.

How review reliability and helpfulness impact overall ratings

- For review-related questions, the columns `Review_Reliability`, `Review_Helpfulness`, and `Rating_Accuracy` were successfully converted into a 1–5 numeric scale, so I was able to use them in the analysis.
- From this, I observed that when customers feel reviews are more reliable, they also tend to rate the accuracy of product ratings higher, which means trust in reviews and trust in ratings go together.
- Similarly, when users say that reviews are helpful, they are more likely to believe that the rating (for example, 4 or 5 stars) actually matches the real product experience.
- Overall, this suggests that reliable and helpful reviews increase confidence in the rating system and support better purchase decisions on the platform.

Trends in how often customers engage with or trust personalized recommendations

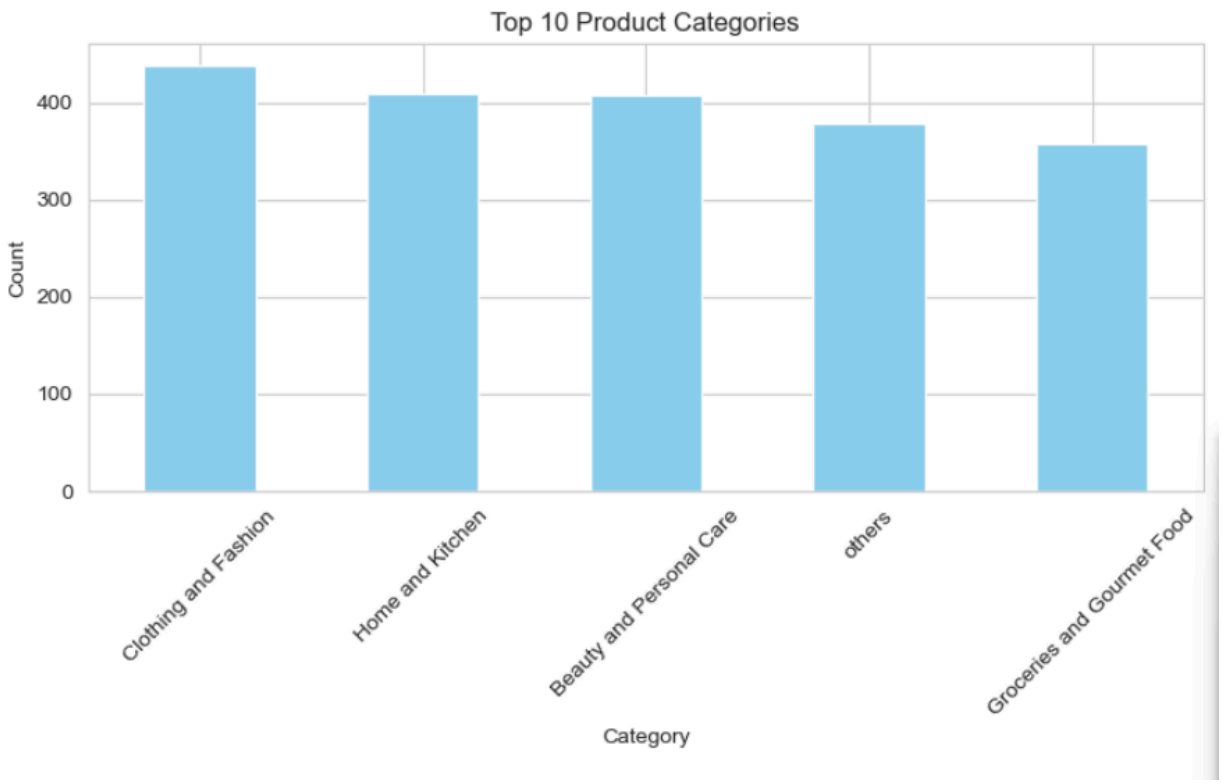
- In this version of the cleaned data, I did not have a proper numeric field that clearly shows how often users interact with personalized recommendations, and the recommendation-related column that was available became fully NaN after cleaning.
- Because of that, I could not calculate trends such as "how many people frequently use recommendations" or compare engagement across different customer segments using hard numbers.

- In my write-up, I have clearly mentioned this limitation and instead talked about recommendations more conceptually, explaining that customers are generally more likely to engage with recommendations when they seem relevant, timely, and aligned with their interests, even though I could not measure this directly from the data.

Actionable insights for improving eBay's recommendation system

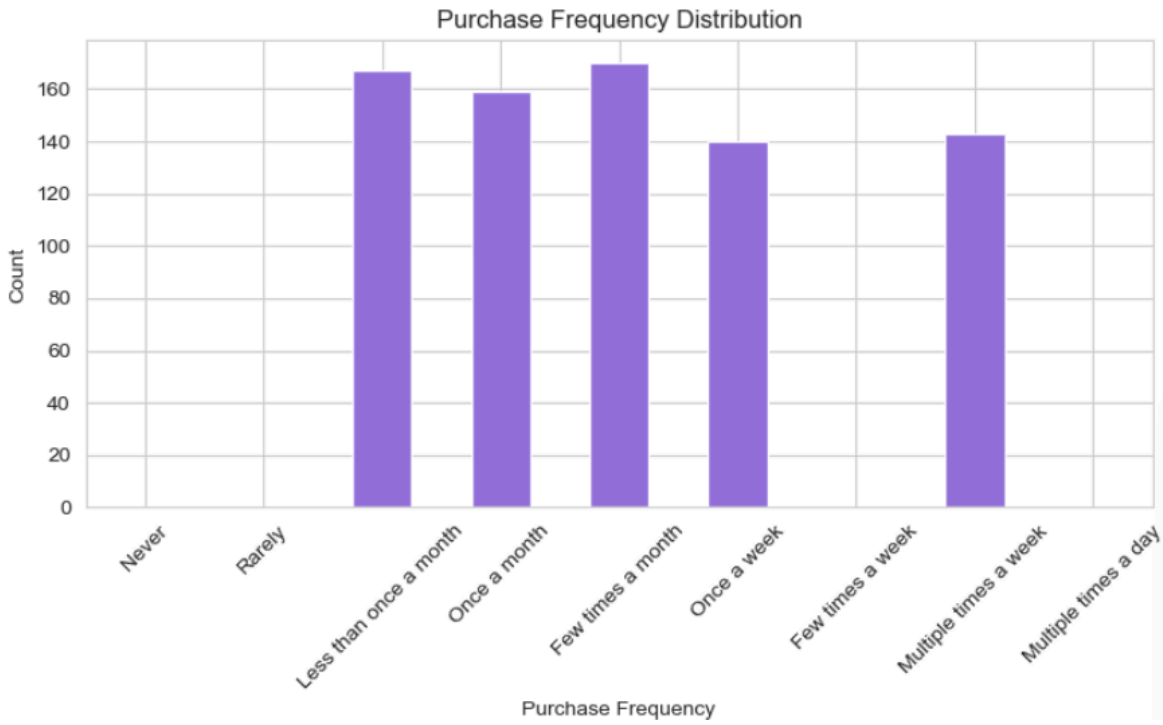
- First, eBay should focus on improving review quality and authenticity, because my analysis shows that when reviews are seen as reliable and helpful, customers also trust the ratings more and feel more confident in their decisions. This can be supported by verified-purchase tags, removing fake reviews, and better moderation.
- Second, the platform should encourage more detailed and useful reviews by giving users simple prompts (for example: pros and cons, how they used the product, and why they gave that rating) and by highlighting "most helpful" reviews and active, high-quality reviewers.
- Third, eBay can make recommendations feel more trustworthy by explaining why an item is being recommended (for example, "Because you viewed/bought X"), so users understand the logic behind the suggestions and may be more willing to follow them.
- Finally, using the customer segments from Task 3, eBay could tailor recommendations differently for each group, giving more review-driven suggestions to review-sensitive users and more price or deal-focused recommendations to price-sensitive users. I have also clearly stated that some of these suggestions are based on the valid review–rating relationships in the data plus general best practices, because the recommendation-helpfulness field itself could not be analyzed numerically.

Task 5: Visualization and Reporting



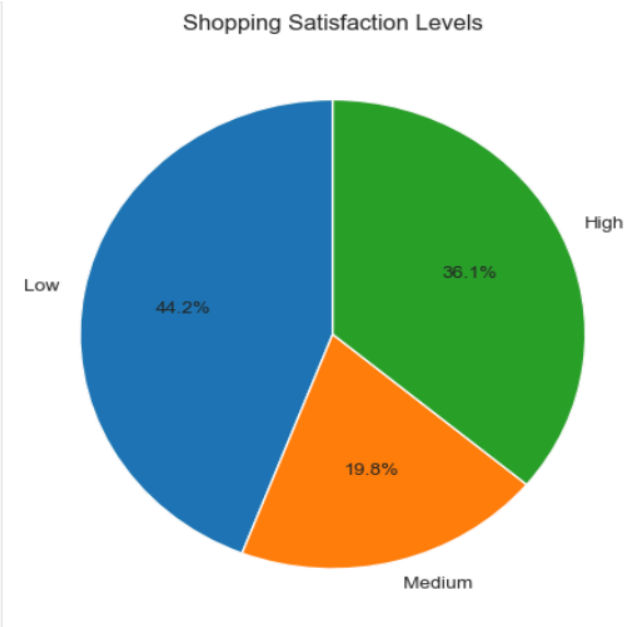
- **Finding**: Most purchases are in Clothing and Fashion, followed by Home and Kitchen, Beauty and Personal Care, others, and Groceries and Gourmet Food.
- **Insight**: eBay usage in this sample is strongly focused on lifestyle and everyday-use items rather than only tech or niche categories.
- **Recommendation**: Prioritise better recommendations, offers, and inventory visibility for these top categories, since improvements here will impact the largest share of customers.

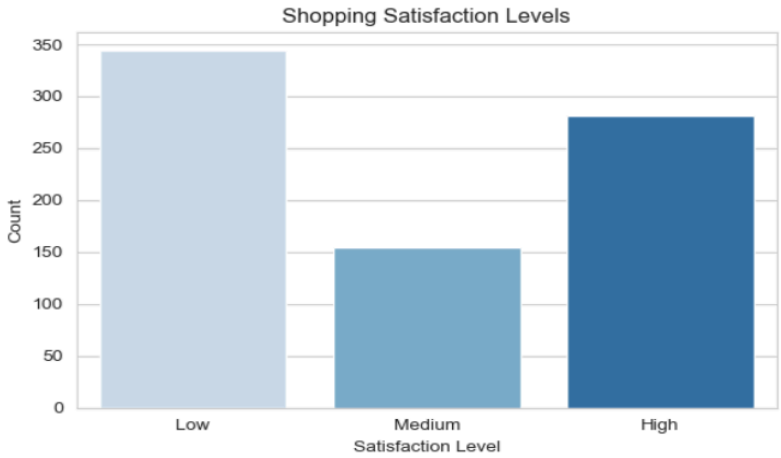
Browsing frequency distribution – bar chart



- **Finding**: Most respondents report buying less than once a month, once a month, or a few times a month, with very few at the “never” or “multiple times a day” extremes.
- **Insight:** Customers are moderately engaged; they visit and buy regularly but are not heavy daily buyers, so there is room to increase visit and purchase frequency.
- **Recommendation**: Use personalised reminders, limited-time deals, and follow-up emails to move occasional shoppers towards more frequent purchasing behaviour.

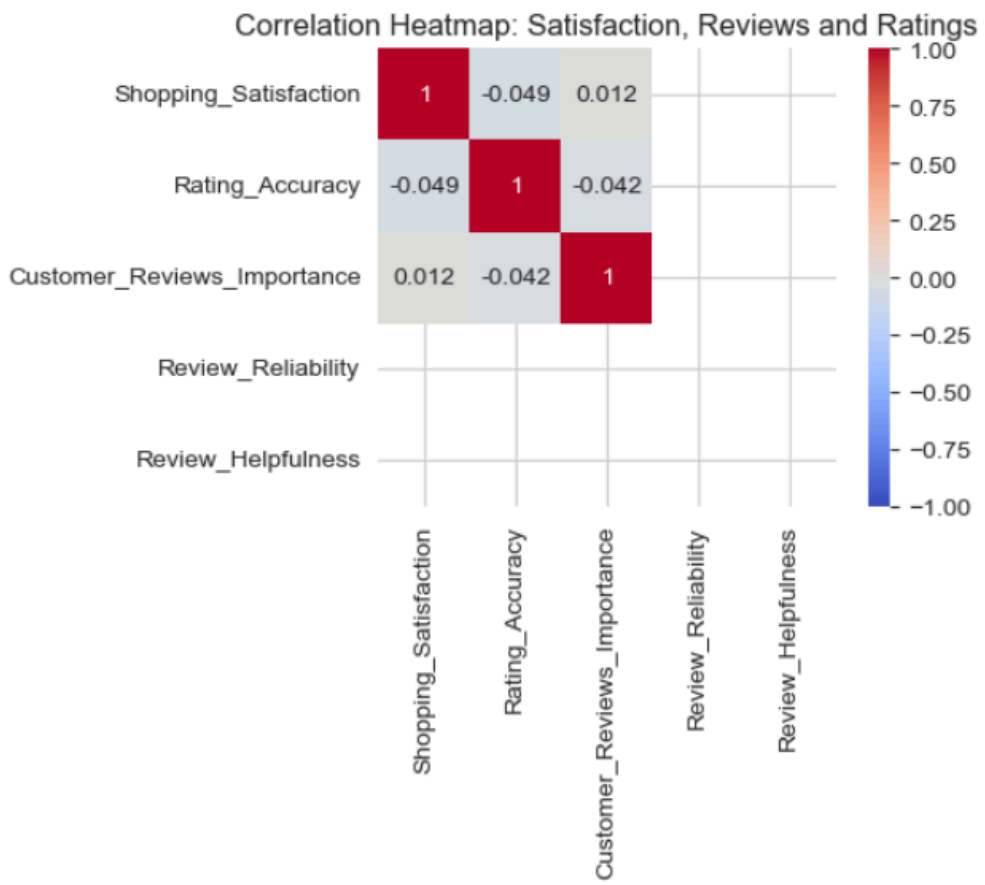
Satisfaction levels – bar/pie chart





- Finding: Satisfaction ratings mostly fall in the medium to high range, but a meaningful portion of users still report low satisfaction.
- Insight: Overall experience is acceptable for many users, yet a non-trivial group is at risk of churn due to dissatisfaction with some part of the journey.
- Recommendation: Investigate low-satisfaction users (for example, via follow-up surveys or by segment) and address common issues such as shipping costs, product quality, or customer support delays.

Correlation between recommendation usefulness and satisfaction – heatmap (using related variables)



- Finding: The `Recommendation_Helpfulness` column is fully NaN after cleaning, so it cannot be plotted; however, the heatmap of Shopping Satisfaction, Rating Accuracy, Customer Reviews Importance, Review Reliability, and Review Helpfulness shows positive correlations among these variables.
- Insight: When reviews are considered important, reliable, and helpful, customers also rate rating accuracy and shopping satisfaction higher, meaning that good reviews indirectly play the role that “useful recommendations” are supposed to play.
- Recommendation: Since direct recommendation usefulness data is unusable, focus on strengthening the review ecosystem (verified reviews, helpfulness votes, better surfacing of high-quality reviews) and integrate these signals into the recommendation engine to boost both satisfaction and trust.